Stony Brook Institute at Anhui University

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Chapter 6
Parameter estiamtion

6.3 Confidence Intervals

- ***One -sample interval estimation of normal distribution***

Introduction

- From the theory of point estimation, we can obtain point estimates for every population parameter.

- Question: How to determine the accuracy of point estimate? 安徽航驰实验室设备有限公司。皖AJ2X20

  For example: We often use sample value $\overline{X}$ to estimate μ , but we do not know how far $\overline{X}$ is from the real value μ because we do not know what μ really is (otherwise we wouldn't need to estimate!).

Introduction

- The solution is the point estimate can be replaced by an entire interval of plausible values. This interval is the Confidence Interval (CI).
- Its advantage is to provide greater confidence than a point estimate.

6.3 Interval estiamtion

- Confidence intervals provide a method of adding more information to an estimator $\hat{\theta}$ when we wish to estimate an unknown parameter θ. An interval estimate consists of a range of values.
- We can find an interval that we think has high probability of containing parameter θ .
- The length of such an interval gives us an idea of how closely we can estimate θ. It is the distance from the lower to the upper end of the interval.

6.3 Interval estiamtion

- The degree of plausibility is specified by a confidence level, such as 95% or 99%.

- Among all confidence intervals with the same confidence level, choose the one that is expected to have the smallest length since it will be the most informative.

6.3 Interval estiamtion

Focus: bilateral interval estimation of a single normal population

Normal population $X \sim N(\mu, \sigma^2)$, Samples $X_1, \cdots, X_n$;

confidence level$1-\alpha$.

Interval Estimation of the Mean of Normal Population

3 Questions:

(1) Find the confidence interval of μ (when σ^2 is known);

(2) Find the confidence interval of μ (when σ^2 is unknown);

(3) Find the confidence interval of σ^2.

Interval Estimation of Normal Population Variance

1. Normal population with known σ

Theorem 6.3.1 Consider a random sample $X_1, X_2, \ldots, X_n$ from $N(\mu, \sigma^2)$, where σ is known. The sample mean $\bar{X} \sim N(\mu, \sigma^2/n)$. Therefore,

$$Z = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0,1).$$

Since U_α is the 100(1 − α)-th percentile of the standard normal distribution, so

$$\boldsymbol{P}\left\{\left|\frac{\bar{\boldsymbol{X}} - \mu}{\sigma / \sqrt{\boldsymbol{n}}}\right| \leq \boldsymbol{u}_{\alpha/2}\right\} = 1 - \alpha,$$

1. Normal population with known σ

$$\Rightarrow \quad P\left\{\bar{X} - \frac{\sigma}{\sqrt{n}} u_{\alpha/2} \le \mu \le \bar{X} + \frac{\sigma}{\sqrt{n}} u_{\alpha/2}\right\} = 1 - \alpha,$$

Therefore, 100(1-α)% CI for μ of $N(\mu, \sigma^2)$ when σ is known is:

$$\left(\bar{X} - \frac{\sigma}{\sqrt{n}} u_{\alpha/2}, \ \bar{X} + \frac{\sigma}{\sqrt{n}} u_{\alpha/2}\right).$$

1. Normal population with known σ

Frequently used values:

$1-\alpha$	0.80	0.90	0.95	0.99
$U_{\alpha/2}$	1.28	1.645	1.96	2.58

1. Normal population with known σ

Example 6.3.1 For a random sample, $X_1, X_2, \ldots, X_{25}$ is a random sample from a normal population N(μ,4), suppose $\bar{X}$ is 19.5. Find a 90% confidence interval for μ.

Since: 1- α=0.9, $U_{\alpha/2} = U_{0.05} = 1.645$,

$$\bar{X} \pm \frac{\sigma}{\sqrt{n}} u_{\alpha/2} = 19.5 \pm 1.645 \frac{2}{\sqrt{25}} = 19.5 \pm 0.66$$

Therefore, a 90% confdence interval is (18.84, 20.16).

1. Normal population with known σ

- The observed interval (a, b) (18.84, 20.16) is simply the obsevered value of the random (A,B). It is one of many possible samples that we could have observed, or may yet observe in the future.

1. Normal population with known σ

- 95% (100(1-α)%) meaning: If sample many times, every sample gives a observed interval (a, b). There are about 95% include μ among these observed intervals, and about 5% are not.

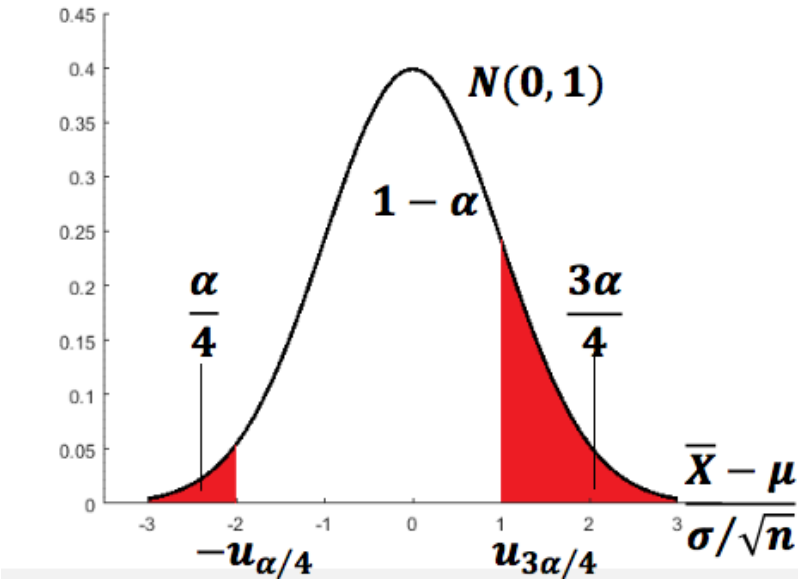
1. Normal population with known σ

Is the CI result unique?

$$P\left(-u_{\alpha/4} < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < u_{3\alpha/4}\right) = 1-\alpha,$$

$$\Rightarrow P\left(\bar{X} - u_{3\alpha/4}\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + u_{\alpha/4}\frac{\sigma}{\sqrt{n}}\right)$$

$$= 1-\alpha.$$



another CI of μ:

$$\left(\bar{X} - u_{3\alpha/4}\frac{\sigma}{\sqrt{n}}, \bar{X} + u_{\alpha/4}\frac{\sigma}{\sqrt{n}}\right).$$

The CI result is not unique!

1. Normal population with known σ

CI

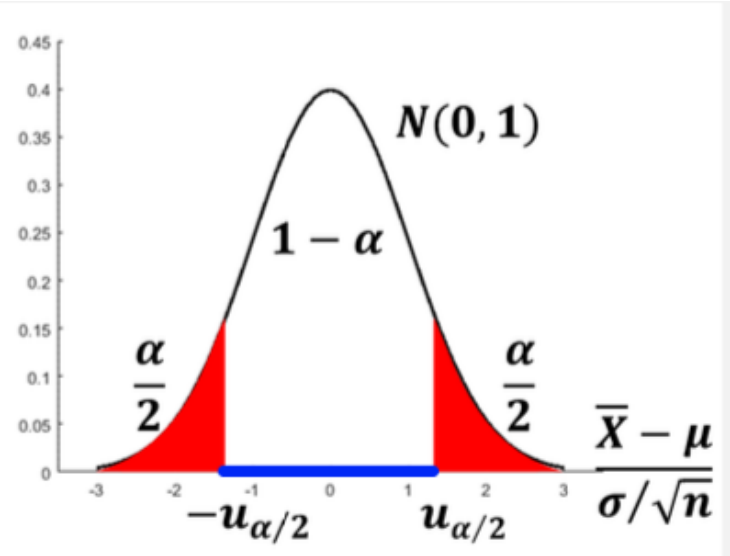
$$\left(\bar{X} - u_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + u_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right).$$

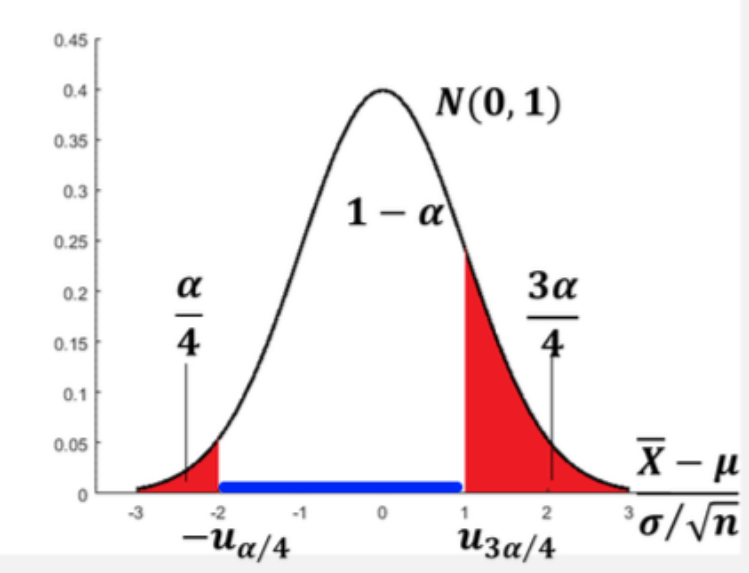
$$\left(\bar{X} - u_{3\alpha/4}\frac{\sigma}{\sqrt{n}}, \bar{X} + u_{\alpha/4}\frac{\sigma}{\sqrt{n}}\right).$$

Length of CI

$$\left[u_{\alpha/2} - \left(-u_{\alpha/2}\right)\right]\frac{\sigma}{\sqrt{n}}.$$

$$\left[u_{\alpha/4} - \left(-u_{3\alpha/4}\right)\right]\frac{\sigma}{\sqrt{n}}.$$

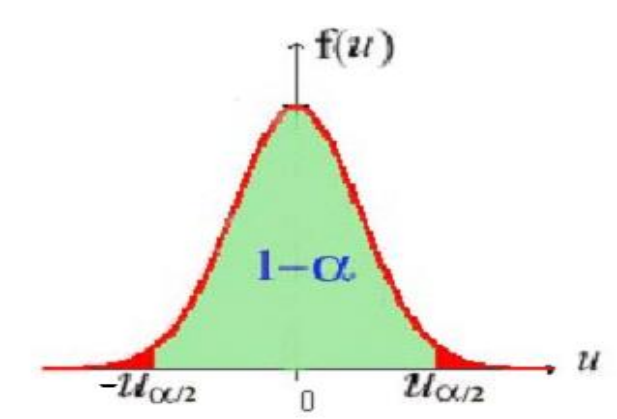




2. Normal population with unknown σ

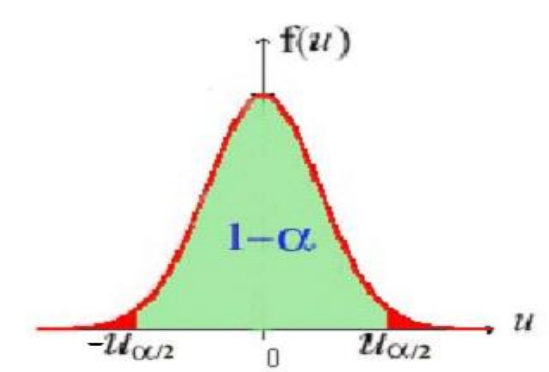
6.3.2 Consider a random sample $X_1, X_2, \ldots, X_n$ is a random sample from a normal population $N(\mu, \sigma^2)$,where σ is unknown. From theorem 5.3.3, we saw that

$$T = \frac{\sqrt{n}(\bar{X} - \mu)}{S} \sim t(n-1).$$



2. Normal population with unknown σ

$$T = \frac{\sqrt{n}(\bar{X} - \mu)}{S} \sim t(n-1).$$



Let $t_\alpha(n)$ be the 100(1 − α)-th percentile of the t distribution with n degrees of freedom. Then

$$P\left\{-t_{\alpha/2}(n-1) \le \frac{\bar{X} - \mu}{S/\sqrt{n}} \le t_{\alpha/2}(n-1)\right\} = 1 - \alpha,$$

2. Normal population with unknown σ

$$P\left\{\bar{X}-\frac{S}{\sqrt{n}}t_{\alpha/2}(n-1)\le\mu\le\bar{X}+\frac{S}{\sqrt{n}}t_{\alpha/2}(n-1)\right\}=1-\alpha,$$

Theorem 6.3.2 Let the sample mean $\bar{X}$ and the sample standard deviation S computed from a small (< 30 n) random sample from a normal population with mean μ, σ^2 unknown. Then a 100 (1−α)% confidence interval for μ is

$$\left(\bar{X}-\frac{S}{\sqrt{n}}t_{\alpha/2}(n-1),\qquad \bar{X}+\frac{S}{\sqrt{n}}t_{\alpha/2}(n-1)\right).$$

2. Normal population with unknown σ

Example 6.3.2 The weights, in pounds, of two-month-old babies in a sample of 15 are the following: 8.9, 8.6, 8.0, 8.3, 8.8, 8.6, 8.1, 7.2, 8.0, 8.6, 9.1, 9.0, 9.1, 8.3, 7.9. The sample mean and sample standard deviation are $\bar{x}$=8.433 and s=0.533, respectively. Assuming that the data were sampled from a normal population distribution, Find the CI of the weight when 1- α=0.95.

Solution: when 1- α=0.95, $t_{\alpha/2}$ (n-1)=$t_{0.025}$ (14)= 2.145

$$\bar{x} \pm \frac{s}{\sqrt{n}} t_{\alpha/2}(n-1) = 8.443 \pm \frac{0.533}{\sqrt{15}} \times 2.145 = 8.443 \pm 0.295$$

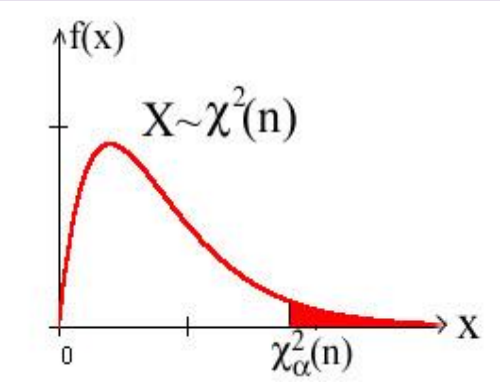
$$= (8.138,\ 8.728)$$

So, a 95% confidence interval for μ is (8.138,8.728).

3. CI for Normal Population Variance σ^2

6.3.3 Consider $X_1, X_2, \ldots, X_n$ be a random sample from $N(\mu, \sigma^2)$, we know that

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1),$$



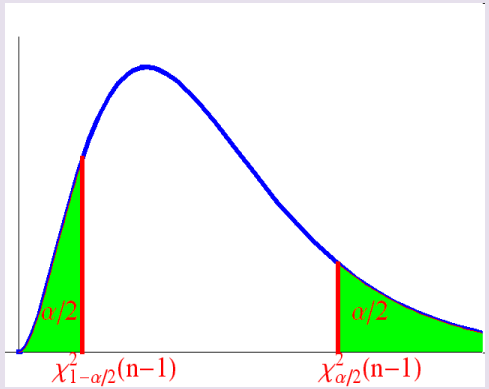
Since $\chi^2_\alpha(n)$ is the 100(1−α)-th percentile of the chi-square distribution with *n* degrees of freedom, so

$$P\left\{\chi^2_{1-\alpha/2}(n-1) < \frac{(n-1)S^2}{\sigma^2} < \chi^2_{\alpha/2}(n-1)\right\} = 1-\alpha,$$

3. CI for Normal Population Variance σ^2

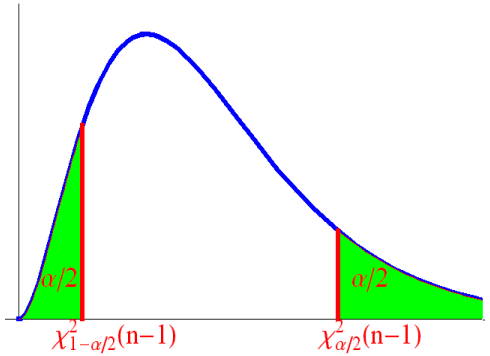
Because the chi-square distribution is not symmetric, the lower bound of the confidence interval is not the negative value of the upper bound. The above probability can be rewritten in terms of σ^2 as follows:

$$P\left\{\frac{(n-1)S^2}{\chi^2_{\alpha/2}(n-1)} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}(n-1)}\right\} = 1-\alpha,$$



3. CI for Normal Population Variance σ^2

$$P\left\{\frac{(n-1)S^2}{\chi^2_{\alpha/2}(n-1)} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}(n-1)}\right\} = 1-\alpha,$$



Thus, a 100(1 − α)% confidence interval for of a normal population is

$$\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2}(n-1)}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}(n-1)}\right).$$

3. CI for Normal Population Variance σ^2

Theorem 6.3.3 Let S^2 be the sample variance computed from a random sample from a normal population N(μ, σ^2). Then a 100(1 − α)% confidence interval for σ^2 is

$$\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2}(n-1)}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}(n-1)} \right).$$

3. CI for Normal Population Variance σ^2

Example : A random sample of 14 eighth-grade boys in Hefei, from a normal population gives the following heights in inches: 60.3, 62.7, 67.3, 62.3, 64.9, 71.2, 59.7, 59.6, 65.5, 68.7, 60.5, 64.2, 65.1, 59.5. Find a 95% confidence interval for the population variance σ^2.

Solution: From the data we get the sample mean 63.7, and the sample variance is $S^2 = 13.562$.

Then a $100(1-\alpha)\%$ confidence interval for σ^2 is

$$\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2}(n-1)}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}(n-1)}\right) = \left(\frac{(n-1)S^2}{\chi^2_{0.025}(13)}, \frac{(n-1)S^2}{\chi^2_{0.975}(13)}\right)$$

$$= \left(\frac{13\times 13.562}{24.736}, \frac{13\times 13.562}{5.009}\right)$$

$$= (7.13, 35.20)$$

Relationship between CI and test

- The confidence interval and hypothesis test have a close relationship. The confidence interval is the same as the acceptance region in a two-sided test.

- So the two-samples interval estimation of normal distribution will be discuss at Chapter 7.3. It is as the same solution method as the one-samples interval estimation of normal distribution.